ENVIRONMENTAL MANAGEMENT · 22ENV231

MOSAIC-Air

Wind-aware urban air pollution forecasting

A spatiotemporal deep-learning pilot for short-term $PM_{2.5}$ forecasts, emerging hotspots, and forecast uncertainty.



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Air risk changes across both time and place

A single citywide number can hide where the next local episode is forming.

01

Uneven monitoring

Sparse station coverage leaves gaps between measured locations.

02

Moving conditions

Wind and weather change how pollution builds and moves through a city.

03

Action needs context

Managers need forecasts with uncertainty and a clear location, not a score alone.

PROJECT QUESTION

Can wind-conditioned station relationships improve multi-horizon $PM_{2.5}$ forecasts and help flag emerging high-pollution episodes?

From monitoring records to decision support

One reproducible pilot links modelling, evaluation, and a location-focused interface.

01

Prepare

Hourly $PM_{2.5}$ + weather
12 station panel

02

Forecast

WAGTF-Net
+1 · +6 · +24 hours

03

Communicate

Hotspots · event probability
P10–P90 uncertainty

Research focus

Benchmark short-horizon forecasts · detect episode risk · test missing-input sensitivity

A documented multi-station benchmark

UCI Beijing Multi-Site Air Quality · historical hourly observations · public CC BY 4.0 dataset

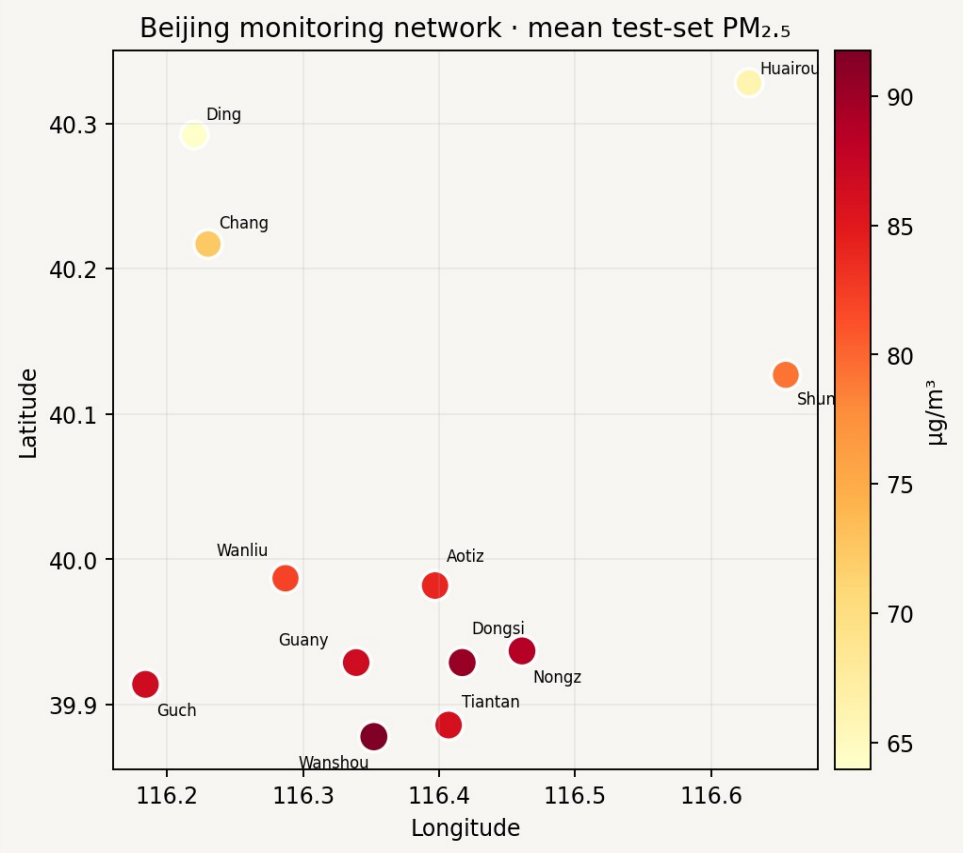
12 monitoring sites

35,064 hourly timestamps

420,768 site-hour records

2013–2017 benchmark period

$PM_{2.5}$ · PM_{10} · SO_2 · NO_2 · CO · O_3
Temperature · pressure · dew point
Precipitation · wind direction · speed



PILOT SCOPE

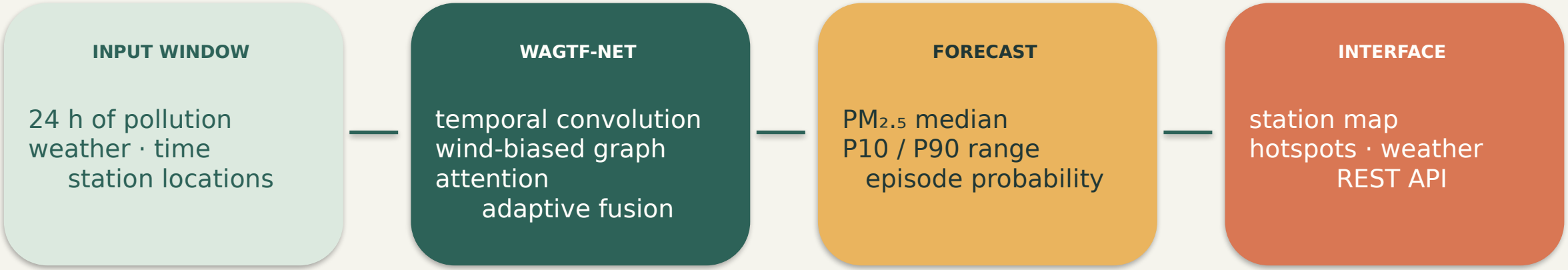
Beijing

Public benchmark makes the method reproducible. It does not represent air quality in Kerala or Amritapuri.

Site coordinates are approximate values added from published station tables.

MOSAIC-Air connects prediction to a readable map

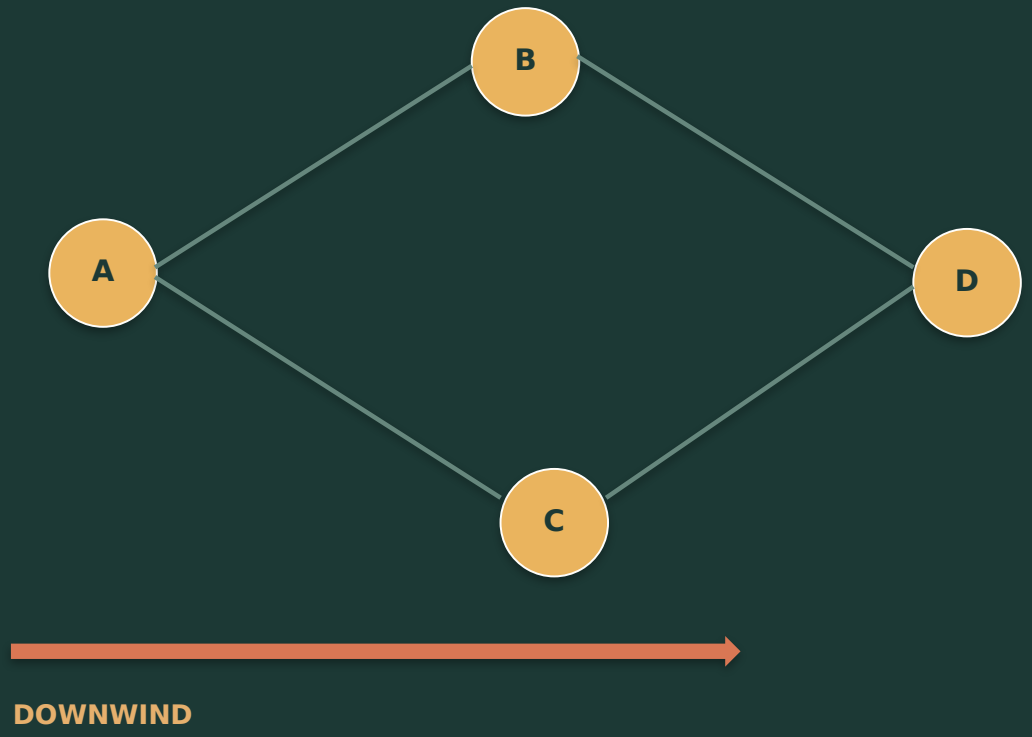
The pilot is implemented end to end: historical panel → model → API → browser dashboard.



Historical replay only Current pollution observations and forecast weather are needed before this pipeline can be used live.

Wind changes how station information is shared

Attention combines learned similarity with direction, speed, and distance.



WAGTF-Net

- 24 h temporal convolution per station
- Graph attention across nearby sites
- Wind alignment biases source → destination influence
- Three horizons × P10 / P50 / P90 + episode probability

Approximate wind-informed attention is a learned influence prior, not a physical transport simulator.

Chronological holdout tests time-forward forecasting

Inputs use the previous 24 hours; issue times are spaced every six hours.

TRAIN

Mar 2013 – Dec 2015

VALIDATE

Jan – Jun 2016

TEST

Jul 2016 – Feb 2017

FORECAST HORIZONS

+1 h | +6 h | +24 h

BASELINES

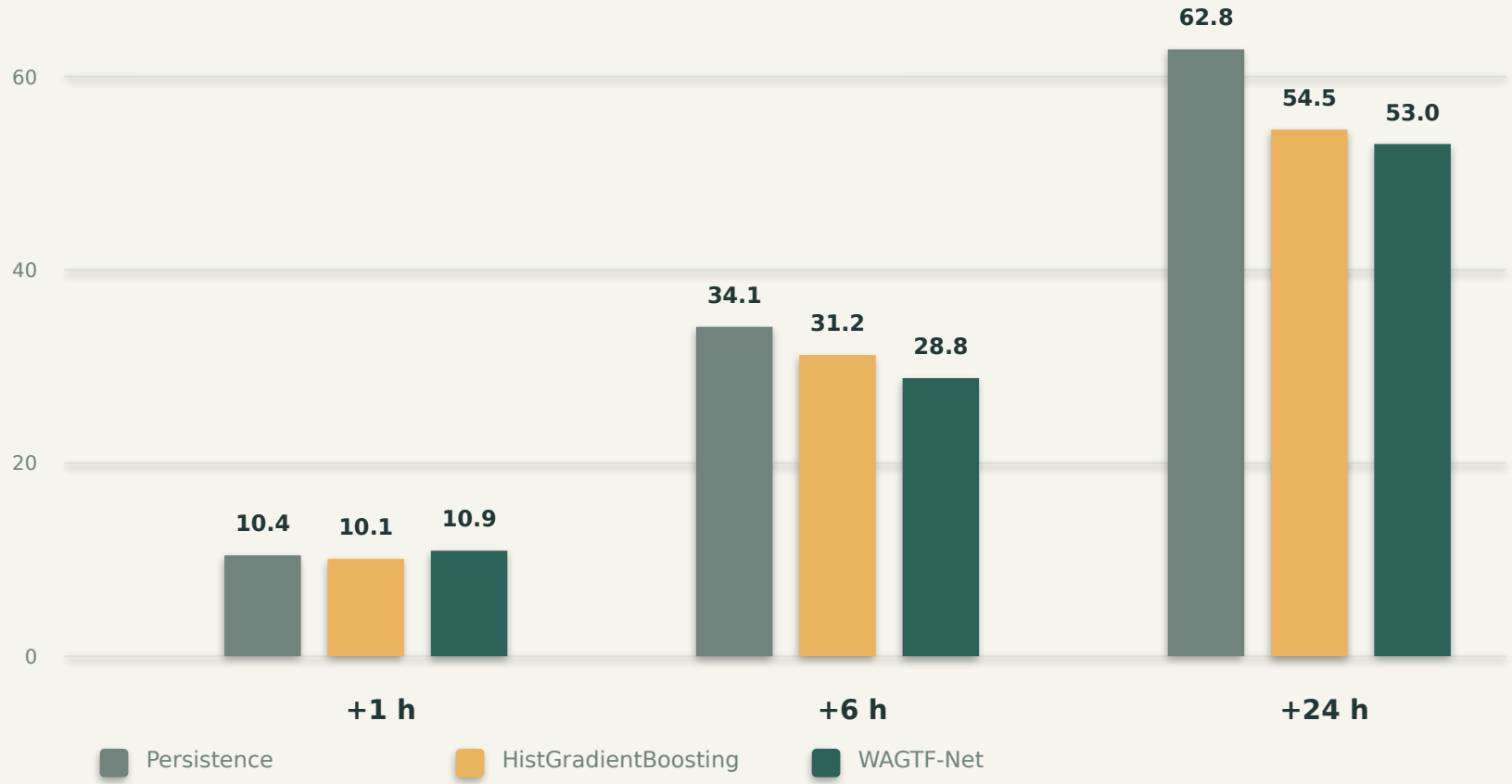
Persistence · HistGradientBoosting

EVALUATION

MAE · RMSE · R² · episode F1 / Brier · P10–P90 coverage

WAGTF-Net is strongest at six hours in this pilot

Held-out MAE (µg/m³) · all event cutoffs selected on validation data



AT +6 HOURS

28.80

µg/m³ MAE

vs 31.19 for Gradient Boosting ≈ 7.7% lower MAE

At +1 h, Gradient Boosting has lower MAE (10.07 vs 10.92).

Uncertainty helps—but missing inputs still matter

Nominal 80% P10–P90 coverage is close overall; the long-horizon interval is broad.

+1 h

82.5% coverage

36.7 µg/m³ mean width

+6 h

84.1% coverage

95.8 µg/m³ mean width

+24 h

80.4% coverage

155.4 µg/m³ mean width

ADDITIONAL 20% $PM_{2.5}$ INPUT MISSING · CHANGE IN TEST MAE

+1 h

+63.6%

10.9 → 17.9 MAE

+6 h

+13.6%

28.8 → 32.7 MAE

+24 h

+0.8%

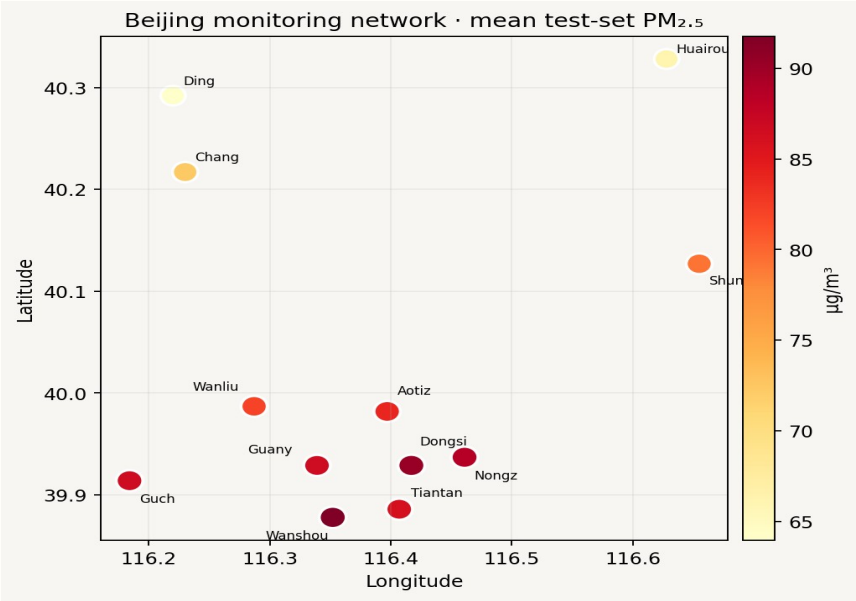
53.0 → 53.5 MAE

Randomly masking extra input cells without retraining. One-hour forecasts degrade most; add missing-data augmentation before deployment.

The dashboard makes forecasts location-specific

A local REST API serves the selected historical issue time to the browser interface.

STATION MAP · GEOGRAPHIC CONTEXT



SELECTED SITE

Station detail

$PM_{2.5}$ median

P50

P10–P90 range
Episode probability
Wind · temperature

INCLUDED

- Interactive station map
- 1 / 6 / 24 h horizon selector
- P10–P90 uncertainty range
- Episode probability + hotspot ranking
- FastAPI JSON endpoints

Historical replay · not live · not a health advisory

What this pilot can—and cannot—say

The result is promising at the medium horizon and bounded by its benchmark design.

WHAT WE FOUND

- 6 h MAE improves about 7.7% over HistGradientBoosting
- 24 h improvement is modest; intervals are wide
- One-hour forecasts depend on recent $PM_{2.5}$ inputs
- Forecasts and hotspot views work end to end

NEXT VALIDATION

- Use a local dataset; test held-out stations and cities
- Join live observations to forecast weather
- Stress-test outages, drift, and systematic missingness
- Calibrate by station/horizon; review thresholds locally

The 75 µg/m³ label is a project benchmark—not a health or regulatory threshold.

A reproducible path from station data to spatial forecasts.

WAGTF-Net shows value at 6 hours in this Beijing case study. The next step is validation with local data, weather forecasts, and environmental stakeholders.

Thank you

